



MICHIGAN ENGINEERING
PROFESSIONAL EDUCATION
UNIVERSITY OF MICHIGAN

Powered By
simplilearn

Professional Certificate in AI and Machine Learning



With dedicated live sessions on the latest AI topics, covering **generative AI, prompt engineering, ChatGPT** and more.

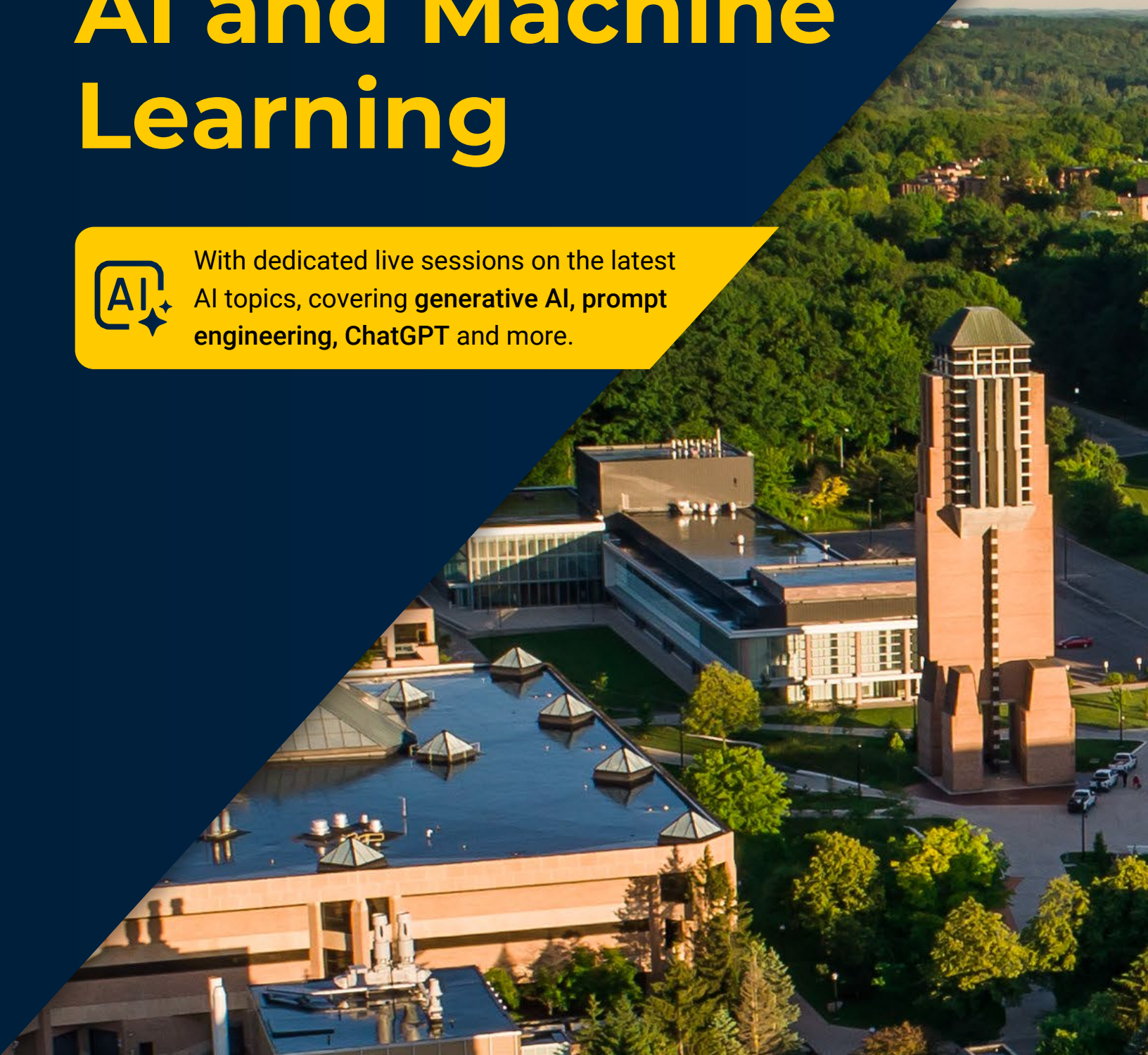


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ABOUT THE PROGRAM

Artificial intelligence (AI) and machine learning (ML) are among the most sought-after and highly compensated digital economy skills. In the past decade, AI has given us self-driving cars, speech recognition, effective web search, and a vastly improved understanding of the human genome. AI is so pervasive that we use it dozens of times daily without realizing it. Many researchers believe it is the best approach to progress towards human-level AI.

Ride this wave and accelerate your career through this comprehensive AI and machine learning program. It provides a comprehensive mix of theory, projects, and extensive hands-on practice while leveraging Michigan Engineering Professional Education's academic excellence. You'll learn the theoretical foundations of AI and ML and how to use the top tools and techniques to tackle real-world business challenges.

This immersive program blends live virtual classes, hands-on projects, labs, and masterclasses to create a high-engagement learning experience. You'll apply AI and ML skills through industry-relevant projects. The curriculum covers essential mathematical and statistical concepts, Python programming, machine learning, deep learning, agentic AI, generative AI, prompt engineering, explainable AI, ChatGPT, natural language processing, and more.

KEY FEATURES



Earn a program completion certificate from Michigan Engineering Professional Education



Learn from a curriculum delivered in live online sessions by industry experts



Attend masterclasses delivered by industry thought leaders and IBM experts



Secure IBM certificates for completing IBM courses



Participate in hackathons and ask-me-anything sessions hosted by IBM



Gain insights into the latest developments in GenAI, prompt engineering, and more



Engage in 12+ hands-on projects, including 3 capstones across various industries



Gain exposure to 15+ tools and libraries like Python, TensorFlow, Numpy, ChatGPT, Claude, Zapier

MICHIGAN ENGINEERING PROFESSIONAL EDUCATION AND **SIMPLILEARN** PARTNERSHIP

The University of Michigan is a top-ranked public university, **#3 among Top U.S. Public Schools, #9 in Artificial Intelligence, and #11 in Best Engineering Schools** (U.S. News & World Report, 2025).

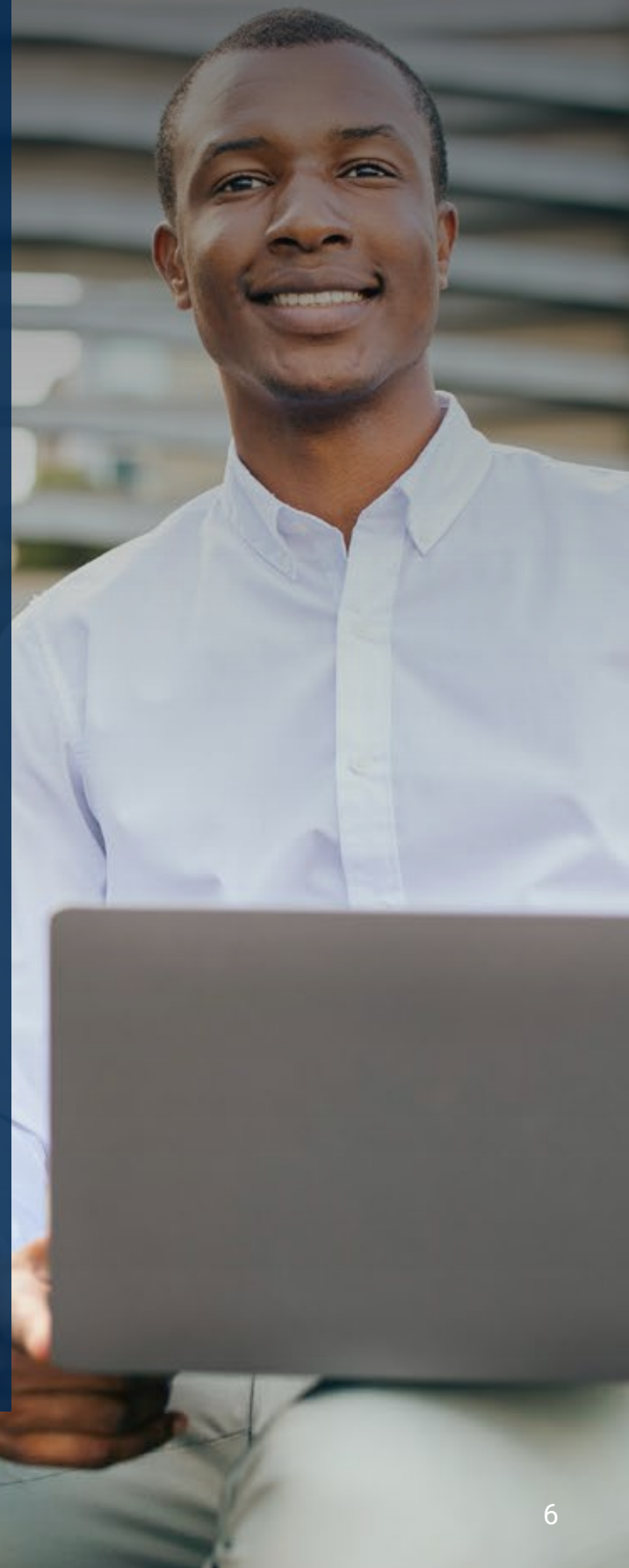
Michigan Engineering Professional Education has chosen Simplilearn to power this program through Simplilearn's award-winning immersive learning model.

The program curriculum is vetted by Michigan Engineering Professional Education, although the instructors are not employees of the University of Michigan. In addition to approving program content, Michigan Engineering Professional Education staff provides oversight of this program in relation to instructor assessment process, quality assurance, student satisfaction, and program outcomes.

WHO SHOULD ENROLL IN THIS PROGRAM?

This program is designed for professionals from diverse industries and backgrounds who want to build skills in AI and machine learning. Whether you're looking to strengthen your technical skills, transition into AI-focused roles, or drive AI-powered projects, this program offers the in-depth knowledge, practical experience, and industry-recognized certificate to help you stay competitive in today's evolving landscape. It is particularly well-suited for professionals with a foundational understanding of programming and mathematics, along with an analytical mindset. This program is ideal for those who want to:

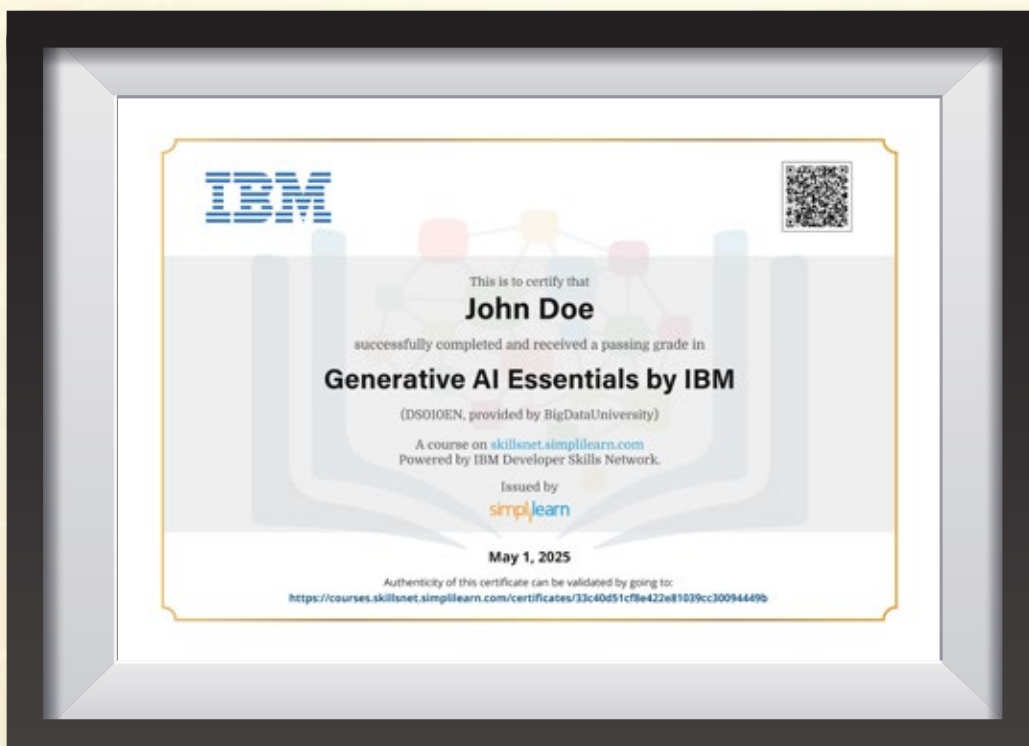
- Improve effectiveness in their current role using AI and ML
- Move into AI and ML roles within their organization
- Advance their career in the tech industry
- Pursue entrepreneurial goals in the AI space



PROGRAM OUTCOMES

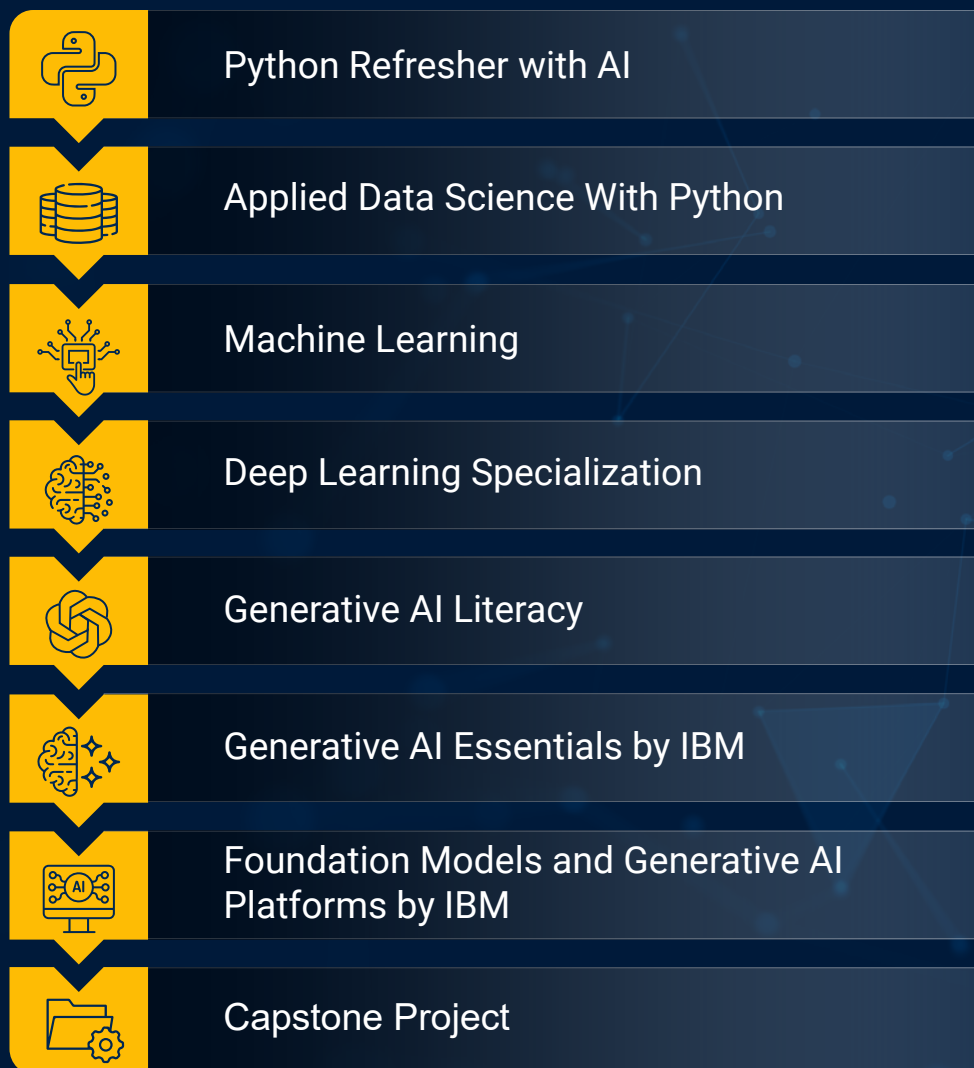
- ✓ Develop the ability to write Python code, utilizing its extensive libraries for data analysis and AI applications.
- ✓ Gain insights into data wrangling, exploration, visualization, and hypothesis testing to enable effective data-driven decision-making.
- ✓ Learn to implement and apply various machine learning algorithms, understanding concepts like regularization, overfitting, and model optimization.
- ✓ Design and train neural networks for complex data interpretations, including convolutional and recurrent architectures.
- ✓ Perform scientific and technical computing using the SciPy package and its sub-packages, such as Integrate, Optimize, Statistics, IO, and Weave.
- ✓ Learn how to create conversational chatbots and customer support agents.
- ✓ Perform text-to-speech conversion with automated speech recognition.
- ✓ Learn about LLMs like ChatGPT, Gemini and their applications in different fields.

COMPLETION CERTIFICATES



LEARNING PATH VISUALIZATION

Core Courses



Electives

- Prompt Engineering Essentials by IBM
- NLP and Speech Recognition
- Advanced Generative AI
- Masterclasses by IBM
- Agentic AI Masterclass

TOOLS COVERED



ChatGPT



Chroma



python[™]
Datetime



GitHub Copilot



gradio



Hugging Face



Imbalanced
learn



jupyter



Keras



LangChain



LabelImg

matplotlib



NumPy



Visual Studio Code



OpenAI API



OpenCV



pandas



PyPDFLoader



python



scikit
learn



SciPy



seaborn



TensorFlow

SKILLS COVERED

- ✓ Generative AI
- ✓ Agentic AI
- ✓ Prompt Engineering
- ✓ Conversational AI
- ✓ Large Language Models (LLMs)
- ✓ ML Model Training and Optimization
- ✓ ML Model Evaluation and Validation
- ✓ Ensemble Learning
- ✓ Deep Learning
- ✓ Natural Language Processing (NLP)
- ✓ Speech Recognition
- ✓ Reinforcement Learning
- ✓ Statistics
- ✓ Machine Learning Algorithms
- ✓ Supervised and Unsupervised Learning



INDUSTRY PROJECTS



Analyzing Customer Orders Using Python

Analyze customer orders with Python data structures to classify products, find buying patterns, and gain insights on popular items, top customers, and category trends.



Building a Python Adventure Game With GitHub Copilot

Build a text-based adventure game using Python and GitHub Copilot to explore, choose paths, and complete a quest while practicing core Python concepts.



Sales Analysis

Analyze Q4 sales data across Australian states to uncover insights that support data-driven decisions for the company's strategy in the upcoming year.



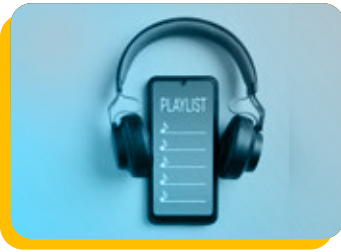
Marketing Campaigns

Conduct exploratory data analysis and hypothesis testing to deepen understanding of the factors influencing customer acquisition.



Employee Turnover Analytics

Create ML programs for HR to check data quality, analyze turnover factors, cluster exits, handle class imbalance, train models, and suggest retention strategies.



Creating Cohorts of Songs

Perform exploratory and cluster analysis to create song cohorts and understand the various factors that contribute to forming each cohort.



Crafting an AI-Powered HR Assistant: A Use Case for Nestle

Create a chatbot that uses PDF data to respond to user queries by extracting text, converting it into vectors, retrieving relevant answers, and building an interface using Gradio.



Creating Designs by Leveraging OpenAI and Gradio UI

Explore OpenAI's API and Gradio UI in creative design through scenarios showing how AI generates images from text, highlighting tech's role in enhancing creativity.



Home Loan Data Analysis

Build a deep learning model using historical data to predict loan defaults, tackling class imbalance and complex features for a safe and secure lending experience.



Lending Club Loan Data Analysis

Design a deep learning model to predict loan defaults for Lending Club, tackling class imbalance and feature complexity for accurate outcomes.



Autonomous Driving

Develop an AI model to detect and segment vehicles in images, then analyze autopilot usage and its impact on road safety as part of this two-stage project.



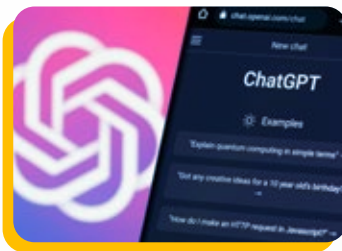
Preserving Heritage: Enhancing Tourism With AI

Design and develop an AI model with TensorFlow to predict structure categories in images, then perform EDA and build a recommendation engine to guide tourist visits.



Sales Forecasting

Forecast item sales across restaurants for Fresh Analytics to understand demand trends over the years, enabling informed decision-making and strategic planning.



ChatGPT-Based Storytelling

Create an interactive storytelling adventure using ChatGPT, where you can create unique and engaging narratives without requiring any coding or additional platforms.



ELIGIBILITY CRITERIA

For admission to this Professional Certificate in AI and Machine Learning program, candidates should:

- ✓ Be at least 18 years old and have a high school diploma, bachelor's degree, or equivalent.
- ✓ Have a basic understanding of programming concepts and mathematics.
- ✓ Have 2+ years of professional work experience (preferred but not mandatory).

APPLICATION PROCESS

The application process consists of three simple steps:



Submit an Application

Complete the application by providing the essential details about yourself.



Reserve Your Seat

Secure your seat by completing the program fee payment.



Start Learning

Begin your learning journey on the designated cohort start date.

TALK TO AN ADMISSIONS COUNSELOR

We have a team of dedicated admissions counselors to guide you in the application process and related matters. They are available to:

- ✓ Answer your questions about the application process.
- ✓ Discuss your financing options and assist with financial aid (if required).
- ✓ Provide insight into the curriculum, program outcomes, and more.

[Inquire Now](#)

[Contact Us | 1-800-212-7688](#)

Python Refresher with AI

Python is the foundational programming language for AI development, making this course a crucial first step in tackling AIML. By mastering Python fundamentals, you will establish a strong technical base to advance into AI model training, deployment, and governance.

Learning Outcomes

- Write and execute Python programs to solve real-world problems
- Utilize data types, operators, and control structures to build efficient code
- Implement functions, loops, and error handling for structured programming
- Handle files for AI and machine learning applications effectively and efficiently
- Develop foundational coding skills required for AI model development

Topics Covered

- ✓ Introduction to Python Programming
- ✓ Python Data Types and Operators
- ✓ Conditional Statements and Loops
- ✓ Error and File Handling
- ✓ Python Functions



Applied Data Science With Python

This course covers essential data science principles, including data preparation, model development, and evaluation. You'll learn key Python concepts like strings, lambda functions, and lists and dive into NumPy, linear algebra, and statistical concepts such as central tendency, dispersion, skewness, covariance, and correlation. Explore hypothesis testing methods like Z-tests, T-tests, and ANOVA. Additionally, explore data manipulation with pandas and enhance your data visualization skills using Matplotlib, Seaborn, Plotly, and Bokeh.

Learning Outcomes

- Develop a comprehensive understanding of the data science process and its components
- Gain proficiency in using Python and its libraries for data science applications
- Understand the use of NumPy and pandas for data manipulation and analysis
- Construct visually appealing and informative graphs using Matplotlib, Seaborn, Plotly, and Bokeh
- Acquire skills in data wrangling and preprocessing techniques

Topics Covered

- | | |
|------------------------------------|---|
| ✓ Introduction to Data Science | ✓ Probability Distributions |
| ✓ Essentials of Python Programming | ✓ Advanced Statistics |
| ✓ Fundamentals of NumPy | ✓ Working With pandas |
| ✓ Linear Algebra | ✓ Data Analysis, Wrangling, and Visualization |
| ✓ Statistics Fundamentals | ✓ End-to-End Statistics Application in Python |

Machine Learning

Explore different types of machine learning and their applications, understand the entire pipeline, and focus on supervised learning with regression models and classification algorithms. Learn unsupervised learning, clustering techniques, and ensemble modeling. Evaluate frameworks like TensorFlow and Keras, and gain hands-on experience with PyTorch to build a recommendation engine.

Learning Outcomes

- Examine various machine learning types and their characteristics
- Explore supervised learning and its applications in real-world scenarios
- Understand overfitting and underfitting concepts and learn detection and prevention methods
- Learn different regression models and their applications
- Examine various ensemble modeling techniques such as bagging, boosting, and stacking

Topics Covered

- ✓ Machine Learning Fundamentals
- ✓ Supervised Learning
- ✓ Regression Models and Applications
- ✓ Classification Models and Applications
- ✓ Unsupervised Learning
- ✓ Ensemble Learning
- ✓ Recommendation Systems

Deep Learning Specialization

This course explores the fundamentals and applications of deep learning, highlighting its distinctions from machine learning. Key topics include neural networks, forward and backward propagation, TensorFlow 2, Keras, performance optimization, model interpretability, CNNs, transfer learning, object detection, RNNs, autoencoders, and PyTorch

Learning Outcomes

- Understand the distinction between deep learning and machine learning
- Learn about the applications of deep learning
- Explore forward propagation and backward propagation in deep neural networks (DNNs)
- Comprehend hyperparameter tuning and model interpretability in deep learning
- Implement dropout and early stopping techniques to improve model performance
- Gain expertise in convolutional neural networks (CNNs) for tasks like object detection

Topics Covered

- | | |
|--|---|
| ✓ Neural Networks and Deep Neural Networks (DNN) | ✓ Convolutional Neural Networks (CNNs) and Object Detection |
| ✓ Forward and Backward Propagation in DNNs | ✓ Transfer Learning for Reusing Pre-Trained Models |
| ✓ TensorFlow 2 and Keras Frameworks | ✓ Recurrent Neural Networks (RNNs) and Autoencoders |
| ✓ Hyperparameter Tuning | ✓ PyTorch for Deep Learning Applications |
| ✓ Performance Optimization Techniques | ✓ Model Interpretability and Explainability |

Generative AI Literacy

After exploring various deep learning specializations, we explore machine learning types and learn to develop generative AI algorithms, including chatbots, LLMs, and AI-driven image and video generation.

Learning Outcomes

- Learn various types of machine learning
- Understand generative AI algorithms
- Learn about chatbots and large language models
- Explore multiple tools available for AI tasks

Topics Covered

- ✓ Transformer Algorithms
- ✓ GANs and VAEs
- ✓ ChatGPT Models
- ✓ Large Language Models (LLMs)
- ✓ Neural Networks
- ✓ Image Generation and Video Generation

Generative AI Essentials by IBM

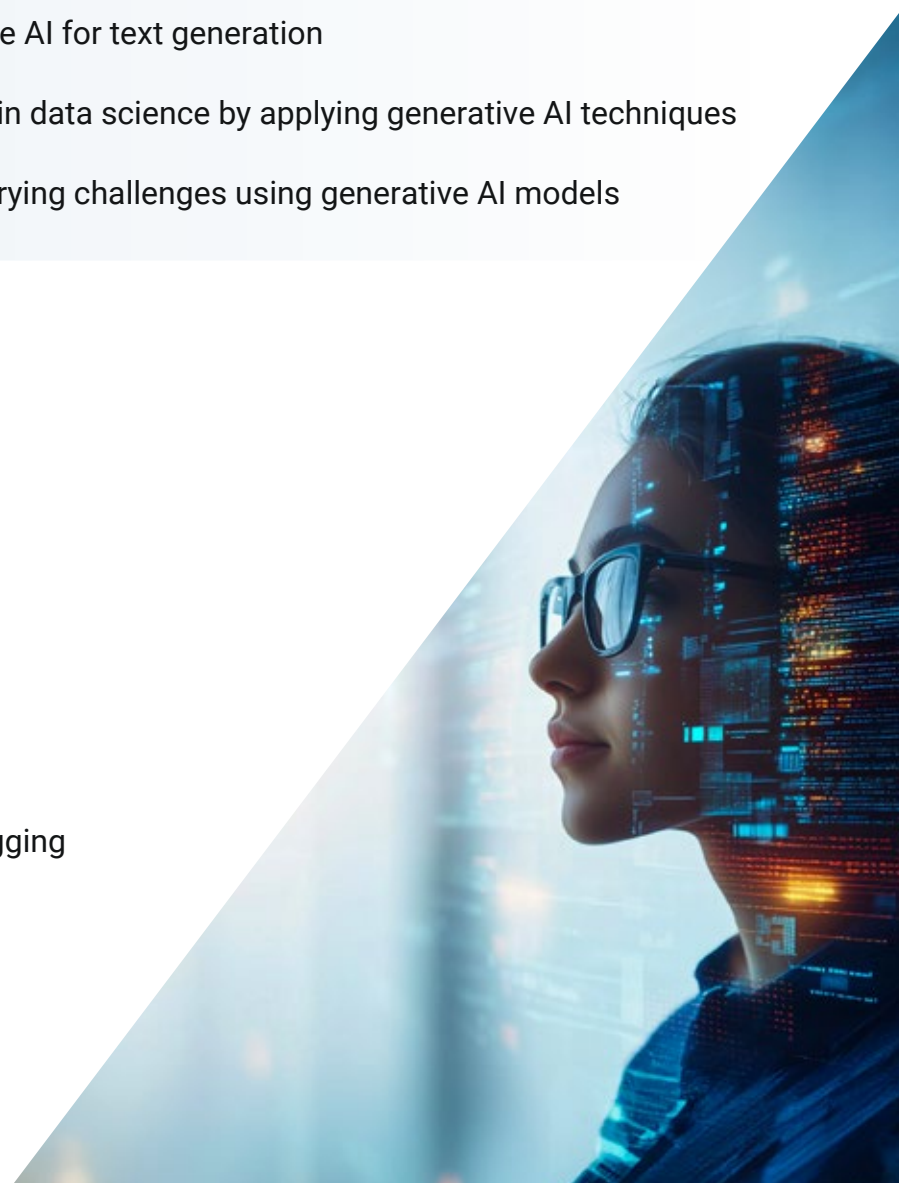
After developing generative AI literacy, we explore the fundamentals, evolution, and applications of GenAI across text, image, audio, video, virtual worlds, code, and data. Learn about popular models and tools like GPT, DALL-E, Stable Diffusion, and Synthesia and their impact across industries.

Learning Outcomes

- Understand the use cases of generative AI for text, image, and code generation
- Describe generative AI and its evolution
- Contrast generative AI with discriminative AI
- Demonstrate use cases of generative AI for text generation
- Enhance skills required for success in data science by applying generative AI techniques
- Overcome data preparation and querying challenges using generative AI models

Topics Covered

- ✓ Artificial Intelligence
- ✓ ChatGPT
- ✓ How To Use Generative AI
- ✓ Learning and Understanding LLMs
- ✓ Natural Language Processing
- ✓ Implementing AI Models Using Hugging Face, TensorFlow, and PyTorch



Foundation Models and Generative AI Platforms by IBM

Upon completing the Generative AI Essentials by IBM module, we move to foundation models, pre-trained AI platforms and their role in text, image, and code generation. Explore IBM Watson, Hugging Face, IBM Granite, GPT, FLAN, and Llama, and gain hands-on experience with generative AI use cases.

Learning Outcomes

- Explore the ability of foundation models to generate text, images, and code using pre-trained models
- Describe the features, capabilities, and applications of IBM watsonx
- Explain Hugging Face as a community for building AI models for everyone
- Demonstrate understanding of the course concepts through a graded quiz and project

Topics Covered

- ✓ Deep Learning and Large Language Models
- ✓ Generative AI Models
- ✓ Foundation Models
- ✓ Pre-Trained Models: Text-to-Text Generation
- ✓ Pre-Trained Models: Text-to-Image Generation
- ✓ Pre-Trained Models: Text-to-Code Generation
- ✓ IBM watsonx.ai: Generative AI for Business
- ✓ Hugging Face: Open Source Generative AI

Generate*

Capstone

The capstone project marks the culmination of your learning journey, providing a chance to apply everything you've learned throughout the program. Designed to reflect real-world industry challenges, the project allows you to demonstrate your technical understanding of AI and your ability to implement it in practical scenarios. With guidance from expert mentors, you'll work on an industry-relevant project—gaining valuable experience and creating a portfolio-worthy project to showcase to potential employers.



Elective Courses

Prompt Engineering Essentials by IBM

Master AI prompting techniques to interact with models like ChatGPT, Gemini, and Claude. Learn prompt engineering frameworks, optimize AI outputs, and apply the Chain-of-Thought (CoT) approach to enhance AI reasoning and accuracy for better problem-solving and applications.

NLP and Speech Recognition

This advanced course explores applying machine learning algorithms to process vast amounts of natural language data. It focuses on natural language understanding, feature engineering, natural language generation, automated speech recognition, speech-to-text conversion, text-to-speech conversion, voice assistance devices, and building Alexa skills. By the end of the course, you will have a deep understanding of the science behind natural language processing and speech recognition, enabling you to develop advanced applications.

Advanced Generative AI

This course explores the creative potential of generative AI. Dive into generative models like variational autoencoders (VAEs) and generative adversarial networks (GANs), as well as large language models (LLMs) and transformer architectures. Learn about attention mechanisms, LangChain workflow design, and advanced prompt engineering. Gain the necessary skills to develop and refine applications with cutting-edge LLMs for specific tasks.

Masterclasses by IBM

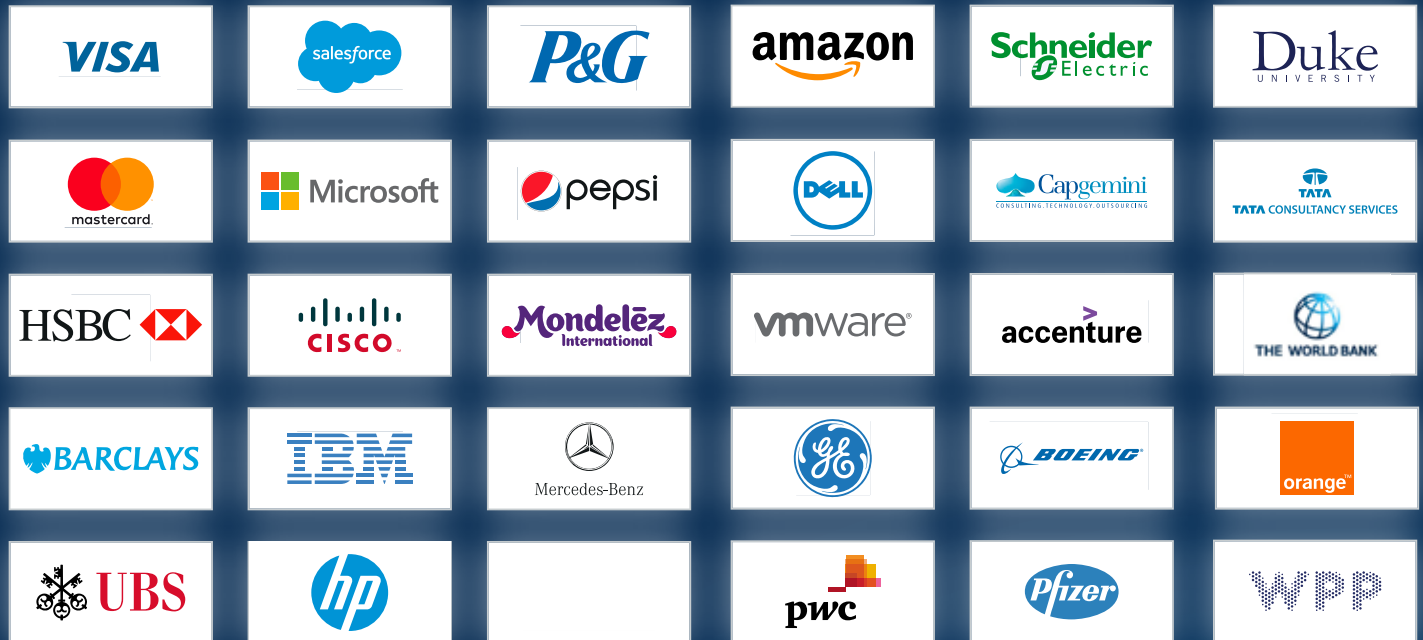
Attend online interactive masterclasses led by experts from IBM. These sessions will equip you with essential skills in artificial intelligence, generative AI, and machine learning, enabling you to drive innovation and maintain a competitive advantage in any industry.

Agentic AI Masterclass

This masterclass will dive deep into the next evolution of AI—autonomous AI agents that can plan, reason, and execute complex tasks with minimal human intervention. Learn how to design, build, and deploy AI-driven agents.



Corporate Training



Features of Corporate Training



Tailored learning solutions



Flexible pricing options



Enterprise-grade learning management system (LMS)



Enterprise dashboards for individuals and teams



24x7 learner assistance and support



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